Chemistry of the Earth: Lecture 4

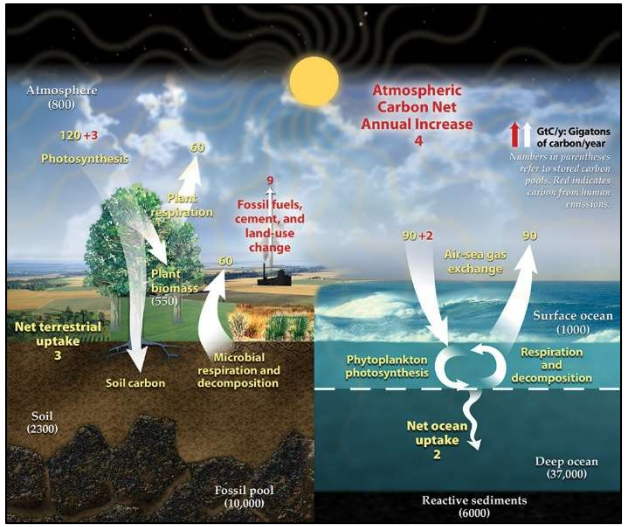
# The Global Carbon Cycle and Organic Chemistry

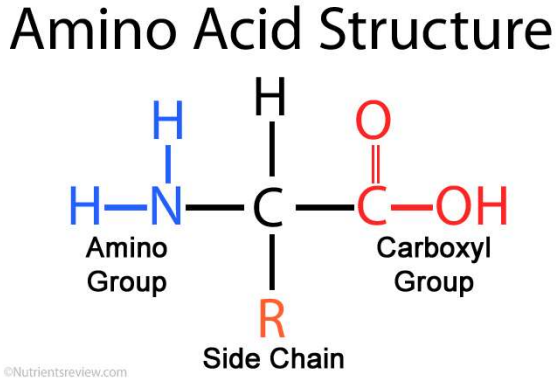


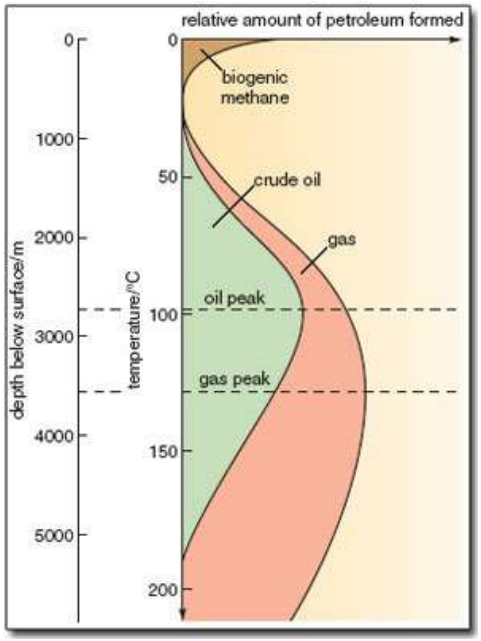
## Part 1: The Carbon Cycle

# OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Introduction to the concept of (bio)geochemical cycles, in particular the global Carbon Cycle
- Review the basics of organic chemistry
- Understand the importance of organic material in the Carbon Cycle



## READING

| | |
|---|---|
| **Grotzinger & Jordan** | **p.355-359, Carbon Cycle** |
| **(basic, minimum)** | **p. 669-670, Biogeochemical cycles, photosynthesis** |
| Marshak | p. 218-219; p. 882-883 |
| **Misra:** | **Ch. 13.6.1 (p. 362-365), Carbon cycle** |
| **Gill:** | **p. 159-162, Organic chemistry** |
| White: | Ch. 12 |
| McSween: | Ch. 6 |

## Further reading, Carbon Cycle:

http://www.carboncycle.org/the-carbon-cycle/
https://www.carboncyclescience.us/what-is-carbon-cycle
http://earthobservatory.nasa.gov/Features/CarbonCycle/?src=eoa-features

**Geochemical Cycles (Biogeochemical Cycles)**

Interactions between:

- Geosphere – the solid Earth (rocks, minerals, soils)
- Hydrosphere – water (oceans, rivers, lakes)
- Biosphere – living organisms
- Atmosphere – gases

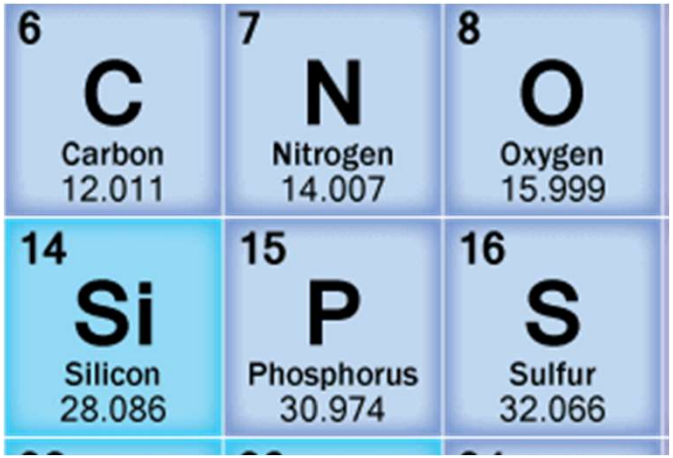


Considerations of

- Global reservoirs
- Transfers between reservoirs

Important cycles:

- Carbon
- Oxygen
- Nitrogen
- Sulphur
- Phosphorus



**The Carbon Cycle**

Importance of the Carbon Cycle:

- C is one of the basic elements for life, ~50% of all living tissues
- $CO_2$ and $CH_4$ are two of the most important greenhouse gases
- Cycling of C approximates the transfer of energy around the Earth, the metabolism of natural, human and industrial systems

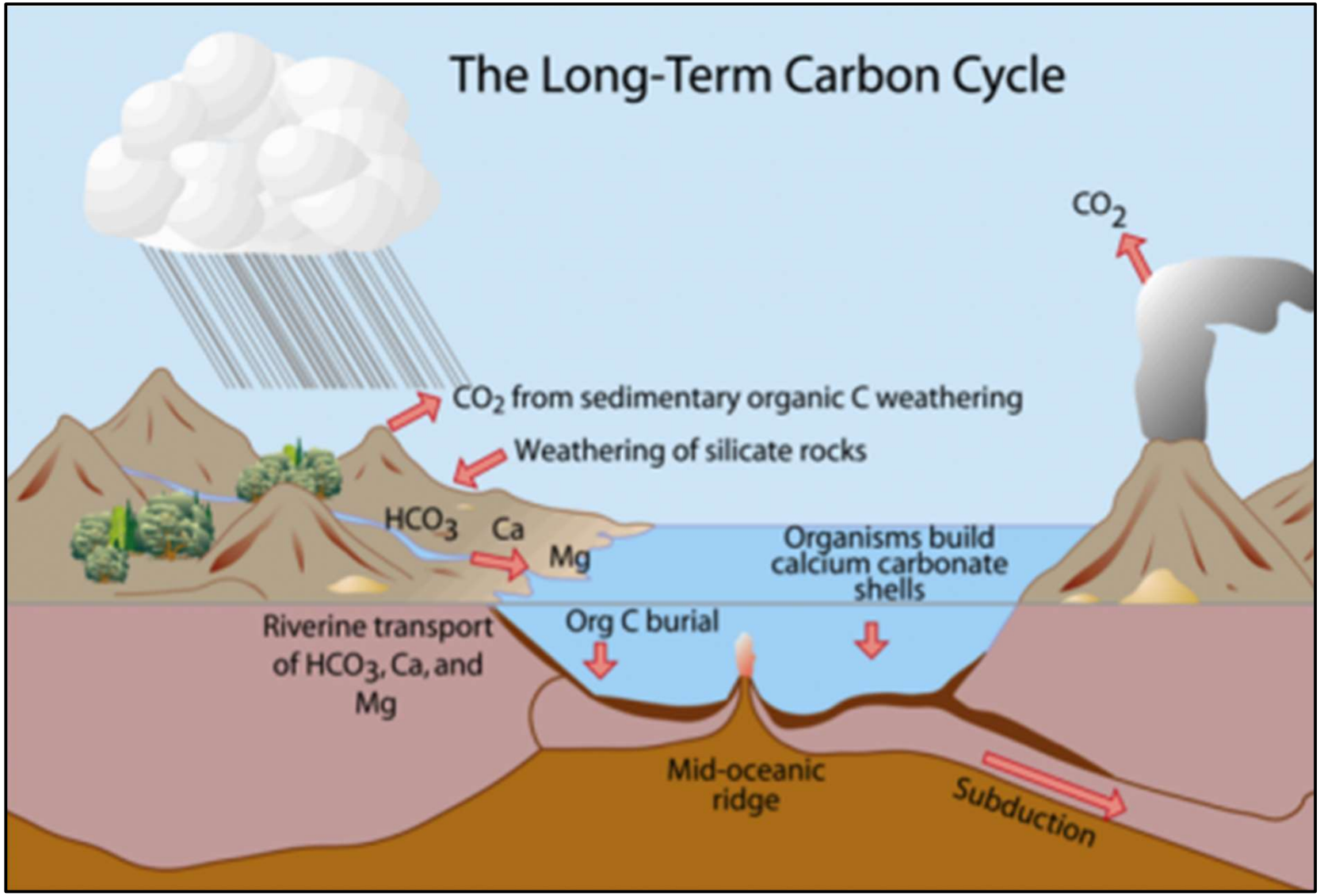


*"The near-surface carbon cycle profoundly affects Earth's changeable climate, the health of ecosystems, the availability of inexpensive energy, and the resilience of the environment.....*

*No other element engages in such an extraordinary range of chemical bonding environments: with oxidation states ranging from −4 to +4, carbon bonds to itself and more than 80 other elements."*
(Hazen and Schiffries, 2013)

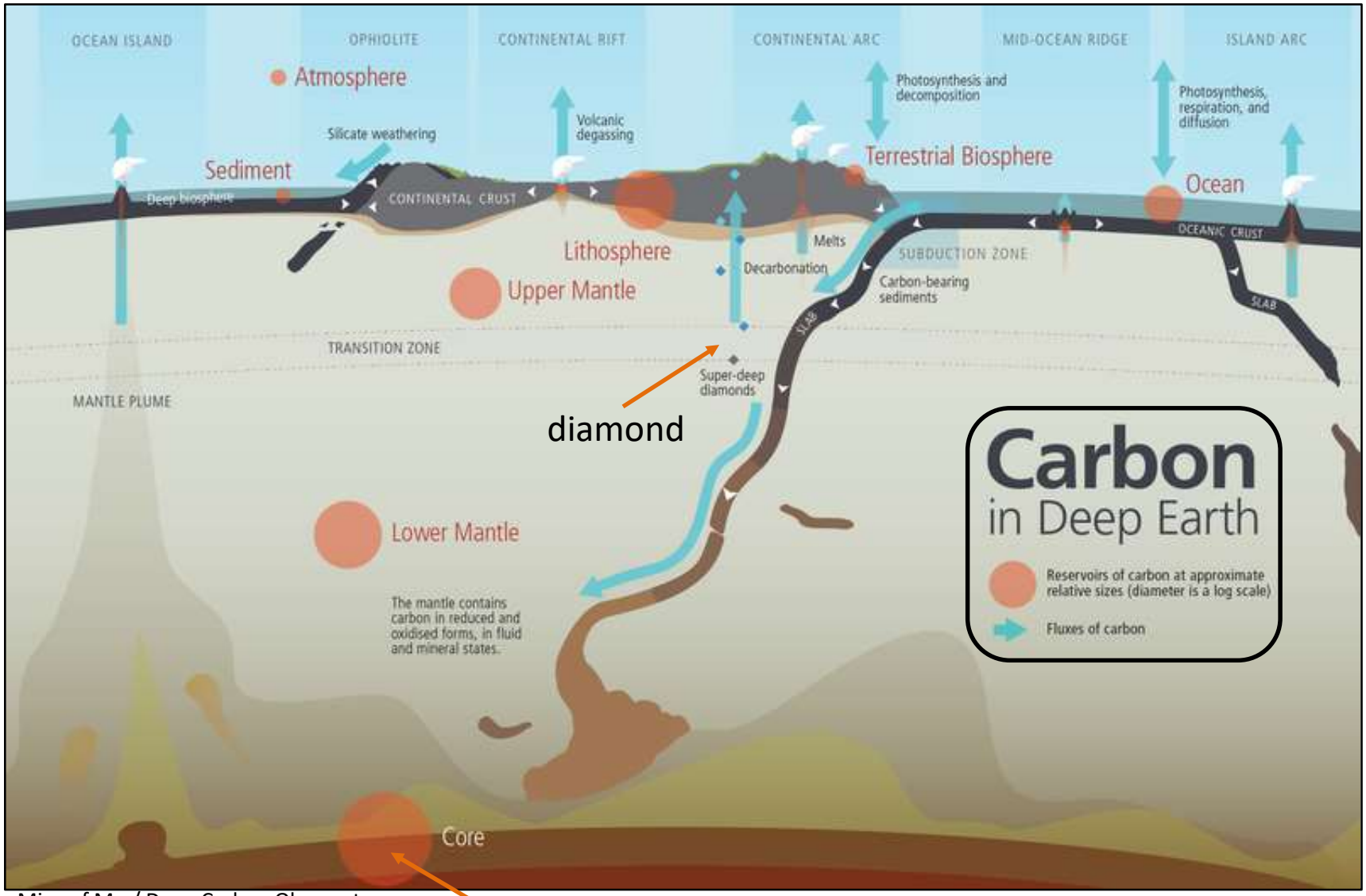
**The Carbon Cycle:**

- Short-term processes involve the atmosphere and hydrosphere
- Long-term processes generally involve rocks




https://uwaterloo.ca/wat-on-earth/news/microbes-minerals-and-isotopes

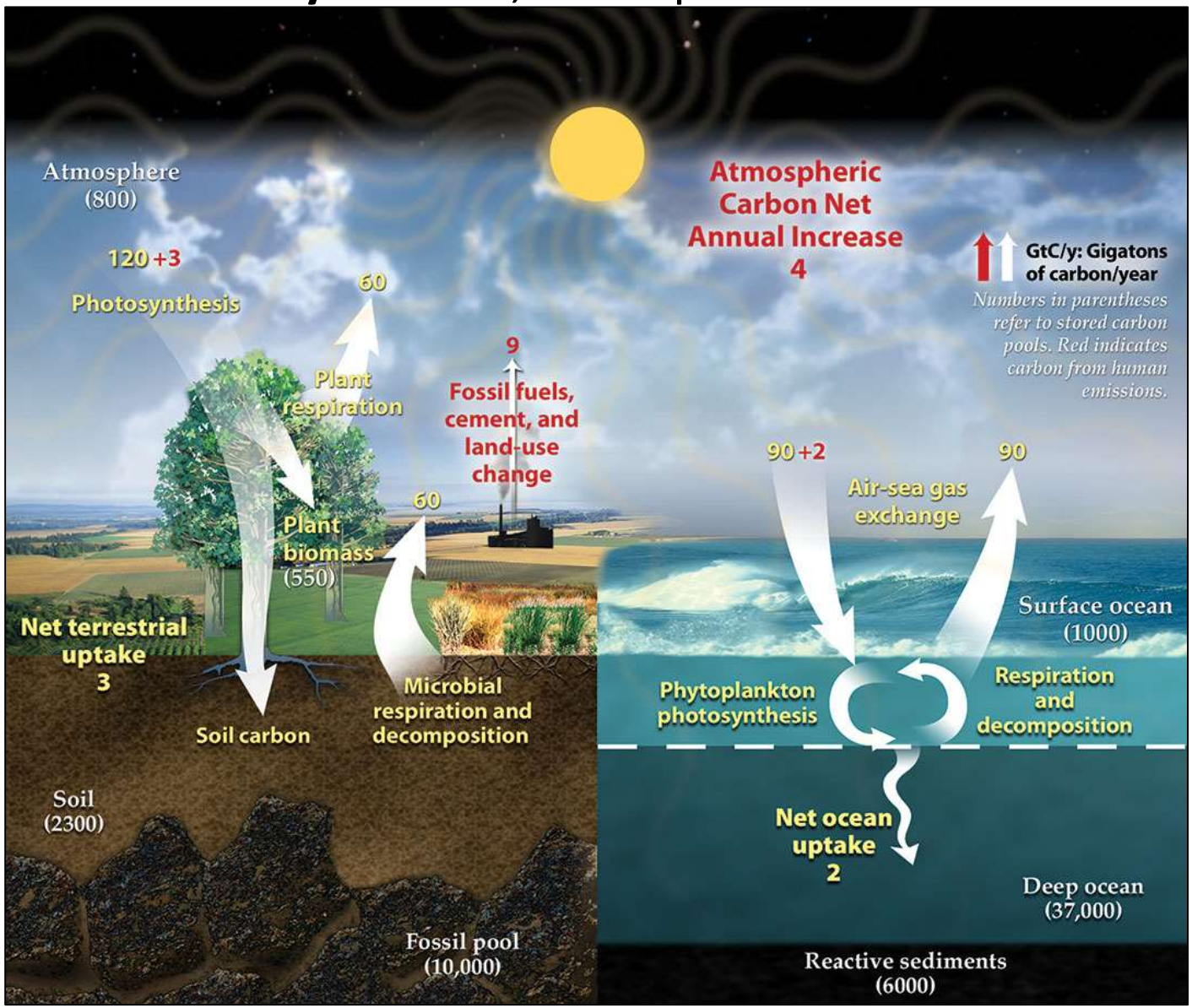
**Deep Carbon Cycle:** Carbon cycles deep into the mantle and core




Mice of Mu / Deep Carbon Observatory

80-90% of the Earth's carbon is in the core

# **Fast carbon cycle:** land, atmosphere and oceans




Yellow: Natural fluxes

Red: Human contributions

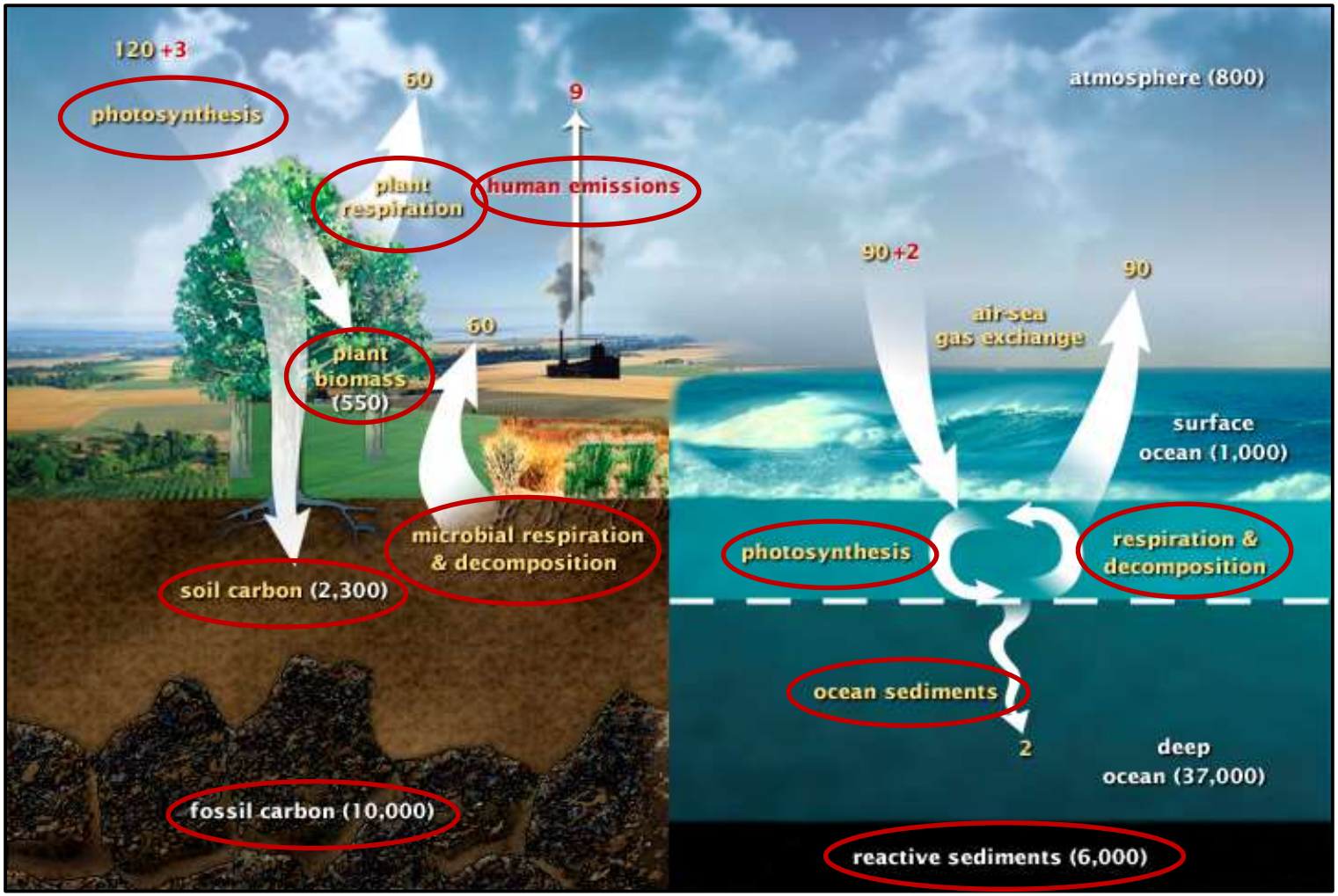
White: Stored carbon, Gt

**The cycle would be balanced without human intervention**

Figure shows movement of carbon between land, atmosphere and oceans in billions of tons per year (GtC/y).

http://www.carboncycle.org/the-carbon-cycle/

**Fast carbon cycle:** land, atmosphere and oceans




**Consider the role of biogenic carbon (carbon related to biomass) in the Carbon Cycle**

**Biogeochemistry:** Study of interactions of the biosphere with Earth systems

**Organic chemistry:** Foundation of biogeochemistry

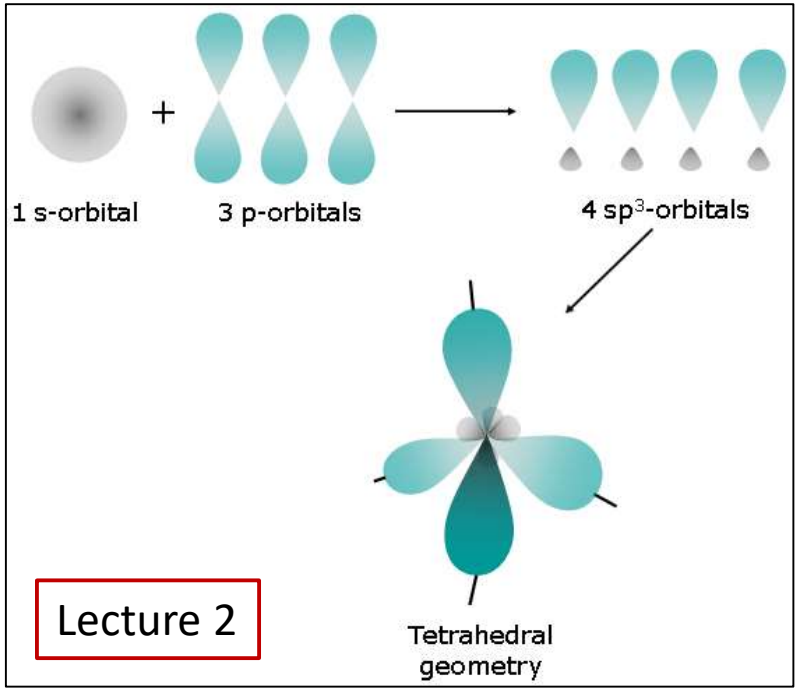
Chemistry of the Earth: Lecture 4

# The Global Carbon Cycle and Organic Chemistry



## Part 2: Introduction to Organic Chemistry

# ORGANIC CHEMISTRY: The Chemistry of Carbon

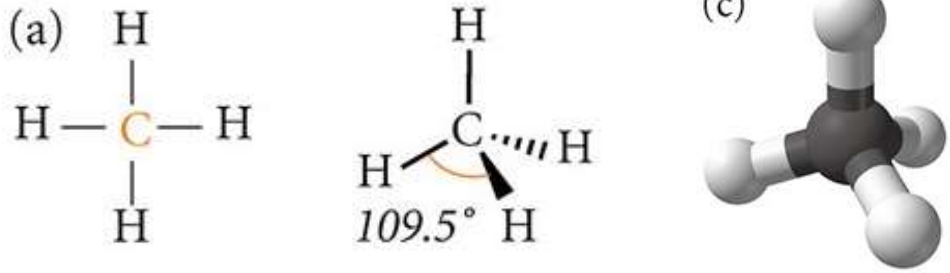



Lecture 2

https://jahschem.wikispaces.com/electronic+structure

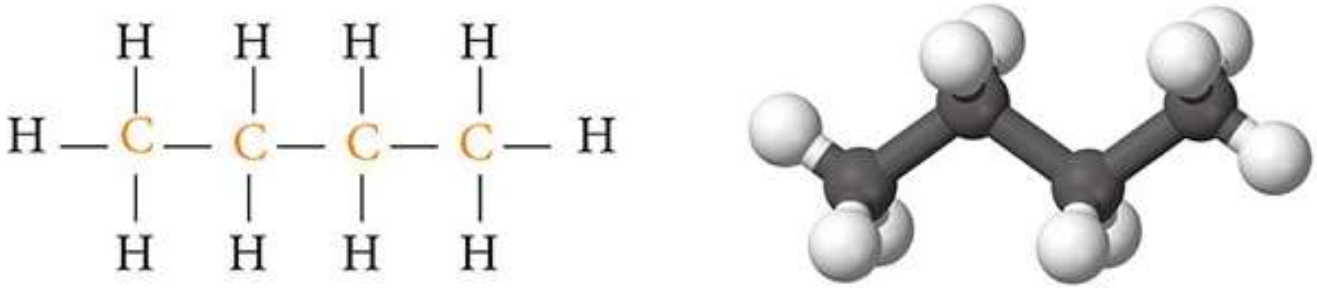
**Chain hydrocarbons (Alkanes)**

- contain only C and H




**Methane ($CH_4$)**

• All bonds are covalent – electrons are shared

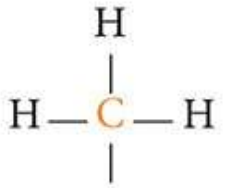


**Butane ($C_4H_{10}$)**

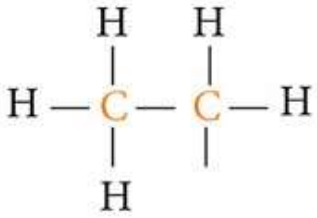
White Fig. 12.2

**Chain hydrocarbons that are attached to something else:**

- Root is called an **alkyl**

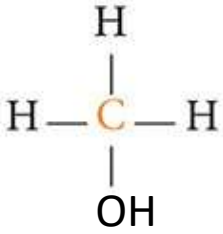


Methyl

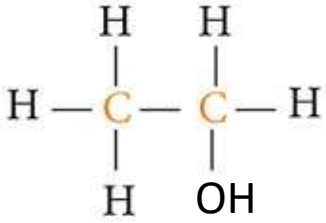


Ethyl

Example:



**Methyl alcohol**

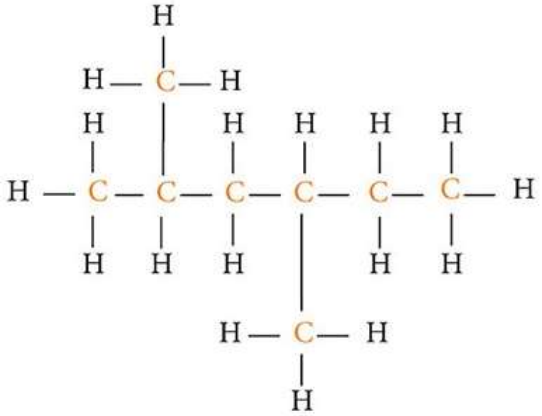


**Ethyl alcohol**

<u>Branched hydrocarbons:</u>

Example: 2,4-dimethylhexane

- Hexane – 6 C atoms
- Dimethyl – two methyl groups attached
- 2,4 means the methyl groups are on the 2nd and 4th C atoms in the chain



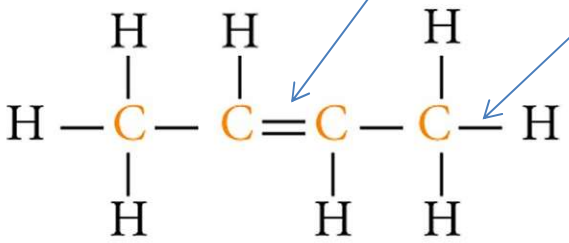
2,4-dimethylhexane

White Fig. 12.2,3

## Double and triple bonds (Alkenes, Alkynes)

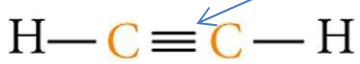
Double bond: 4 electrons shared

Single bond: 2 electrons shared



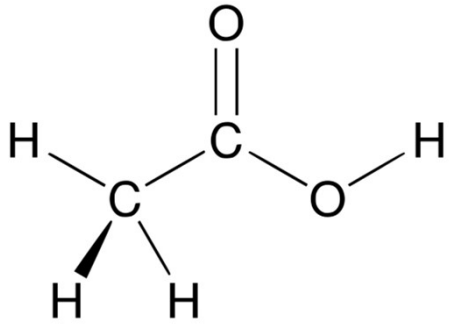
*2-Butene*

Triple bond: 6 electrons shared



*Acetylene*

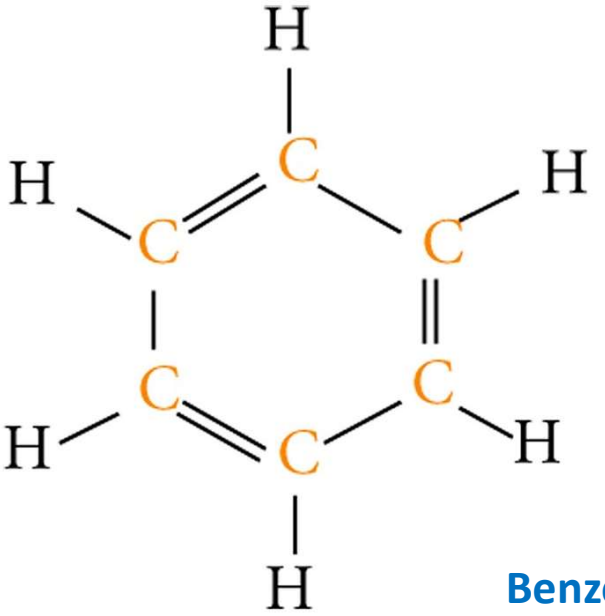
Acetic acid
(Acid or carboxyl group is C=OOH)



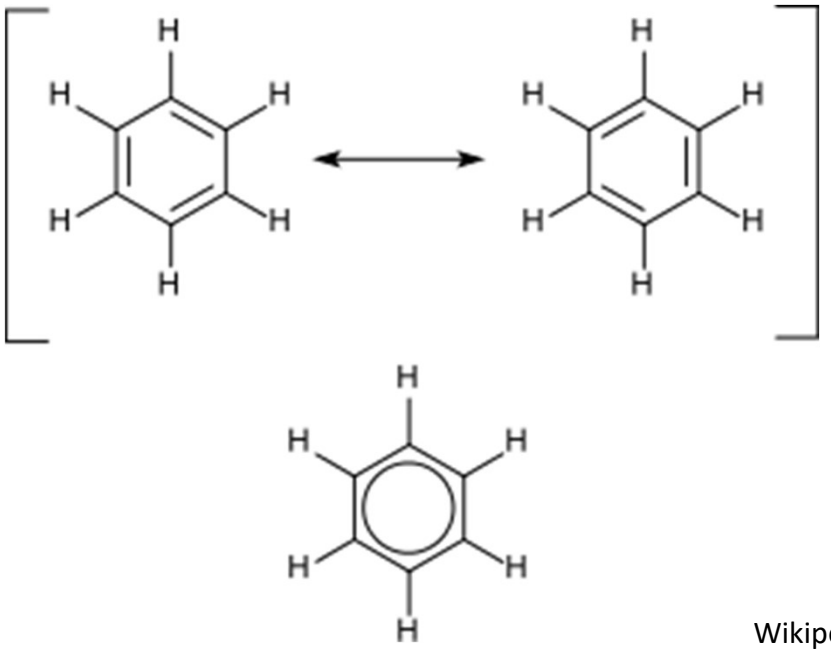
Organic compounds that have chain structures are called **ALIPHATIC.**
Add-ons like -COOH and –OH are called **FUNCTIONAL GROUPS.**

White Fig. 12.4

## Ring Structures – Aromatic compounds



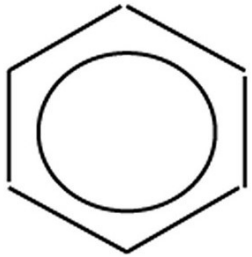
Benzene ($C_6H_6$)



Wikipedia

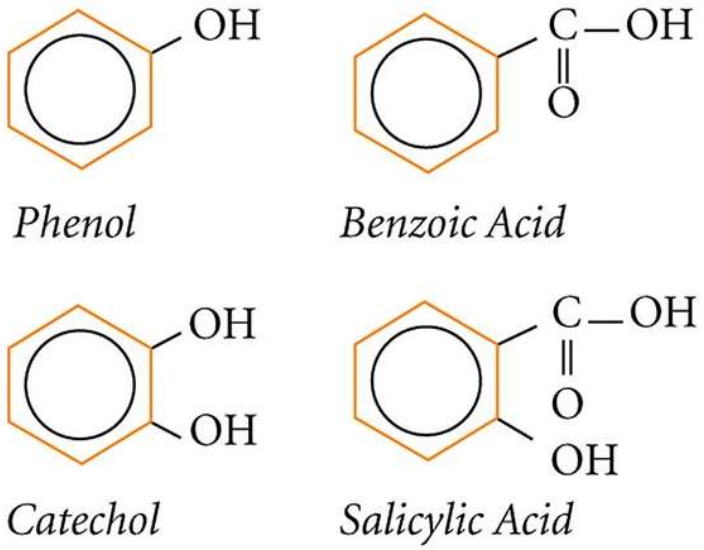
The **benzene** ring:

- Six C atoms form a ring.
- Alternating single and double bonds – electrons become delocalized and spread uniformly around the ring.
- The bonding is written as a circle.



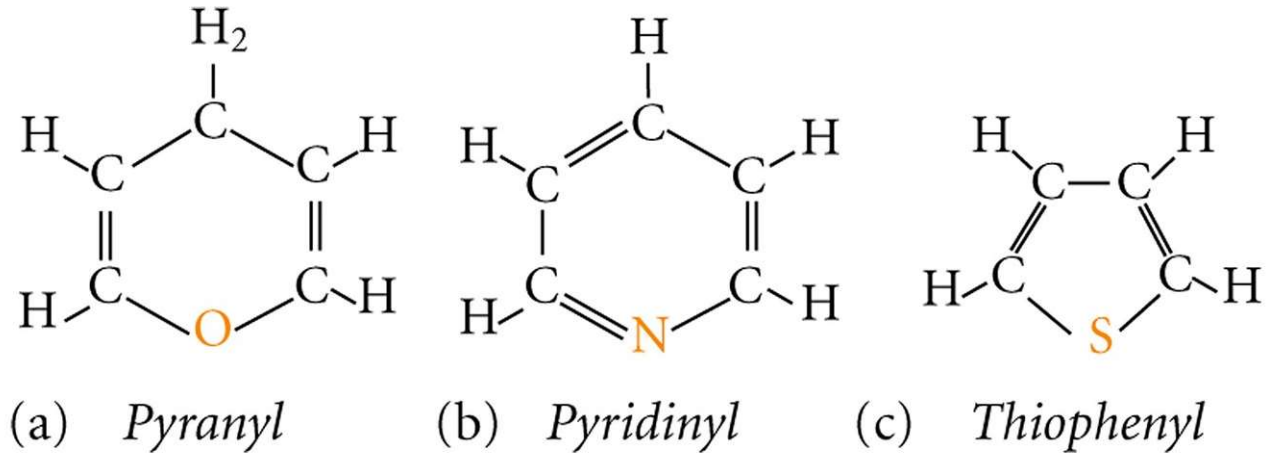
White Fig. 12.5

**Examples of ring structures:**




White Fig. 12.7

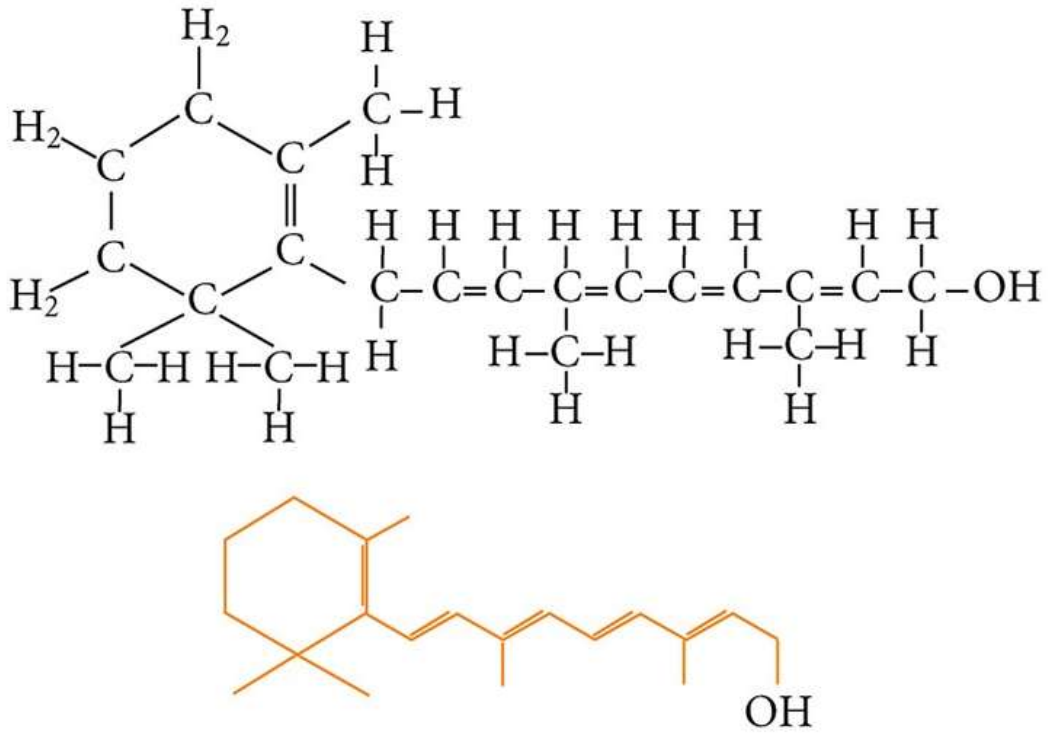
**Other elements can substitute for carbon in the ring structure:**




White Fig. 12.8

**Organic molecules can get complicated so they are represented by a shorthand:**

C atoms are joined by lines. H atoms are assumed to complete the bonding, with the rule that each C atom has 4 bonds.

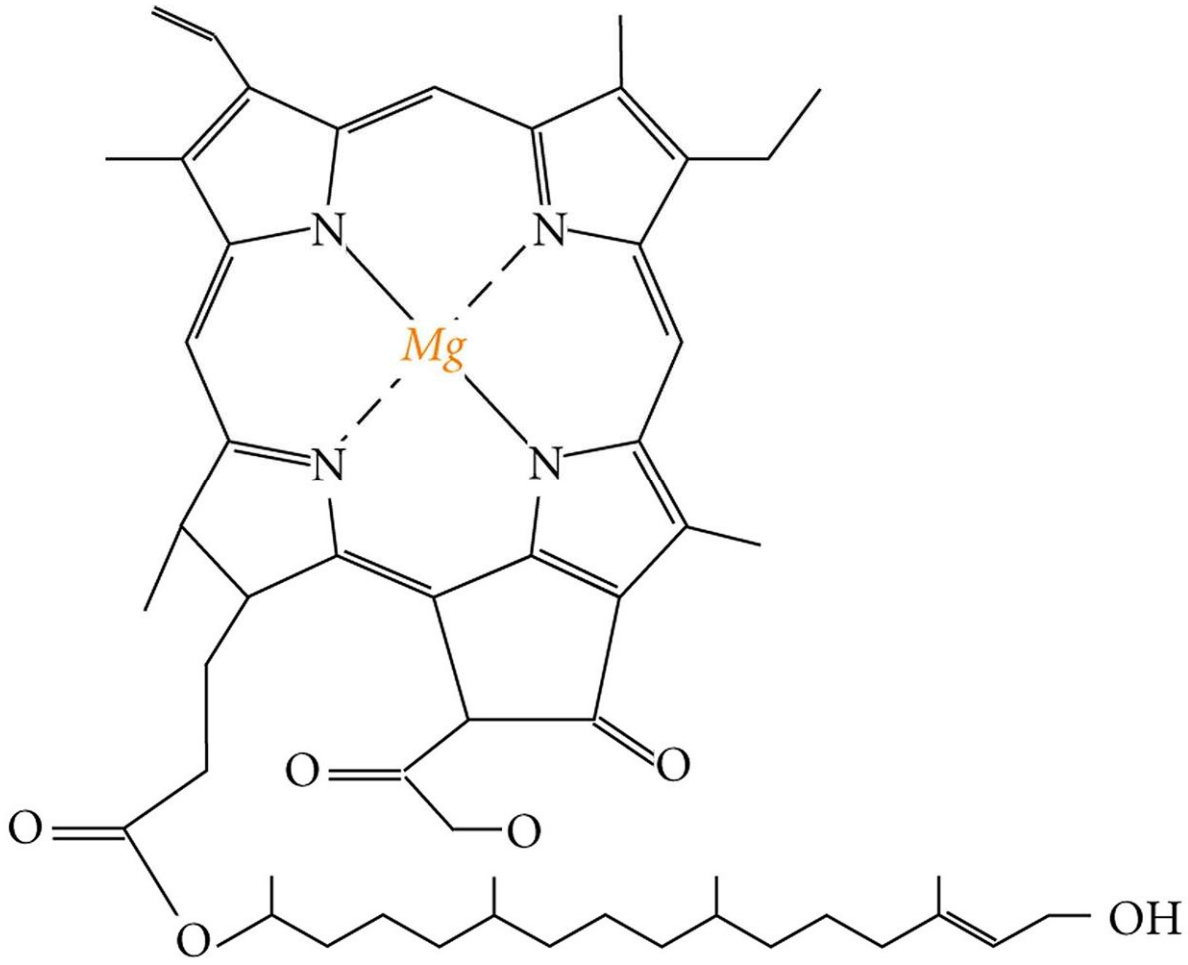



(b) *Retinol (Vitamin A)*

White Fig. 12.9

**Biologically important organic molecules can be very complex.**

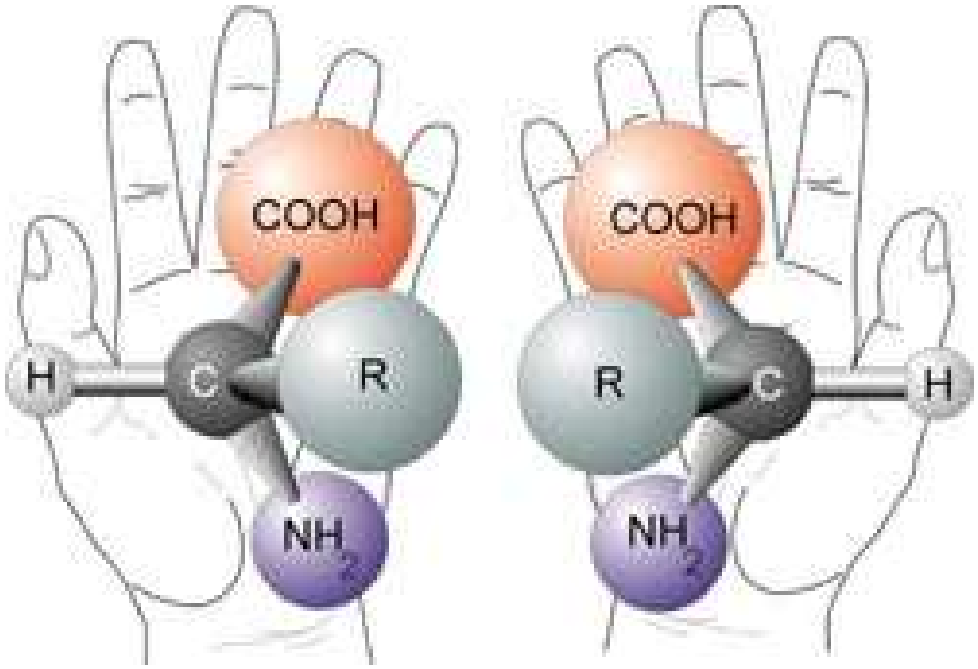
Example: Chlorophyll



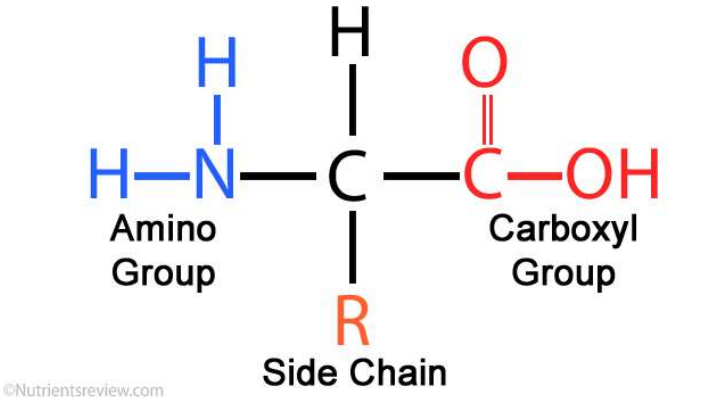

White Fig. 12.15

**Chirality: “handedness”**
**A chiral object cannot be superposed on its mirror image**

In the same way as hands, molecules can be thought of as “left-handed” and “right-handed”



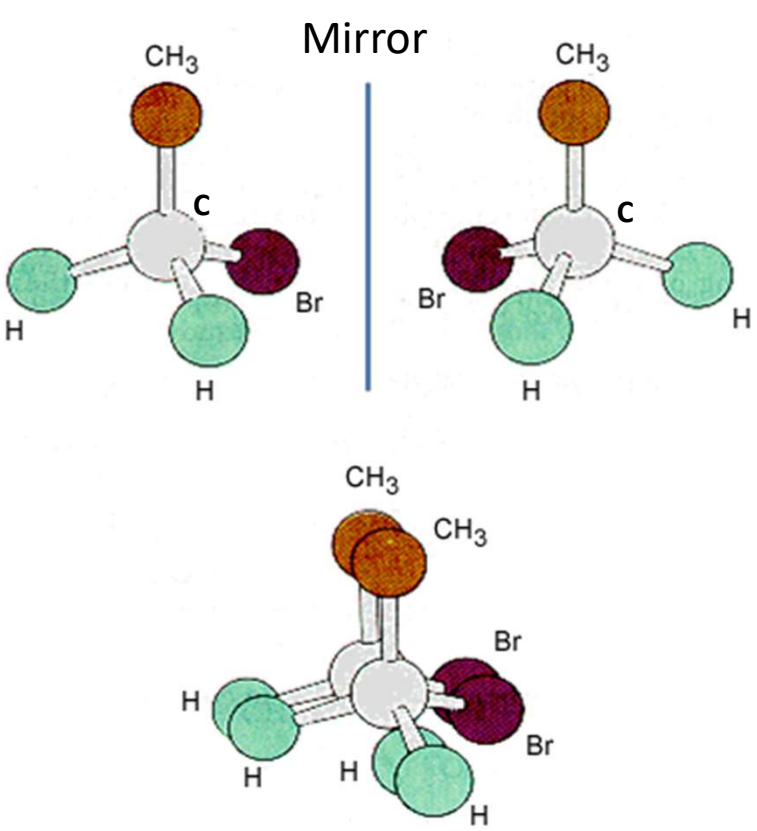

In general, amino acids are chiral molecules



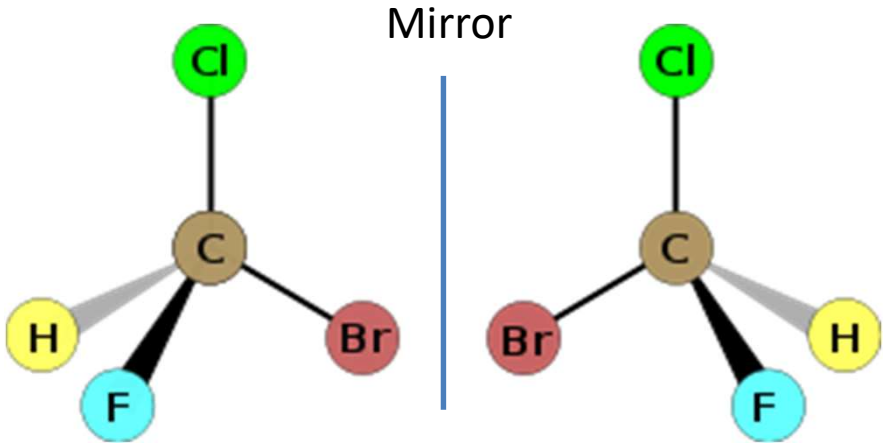

R could be e.g. $CH_3$, or a range of complex carbon chains

https://www.newworldencyclopedia.org/entry/Chirality_(chemistry)

**Chirality: "handedness"**
**A chiral object cannot be superposed on its mirror image**



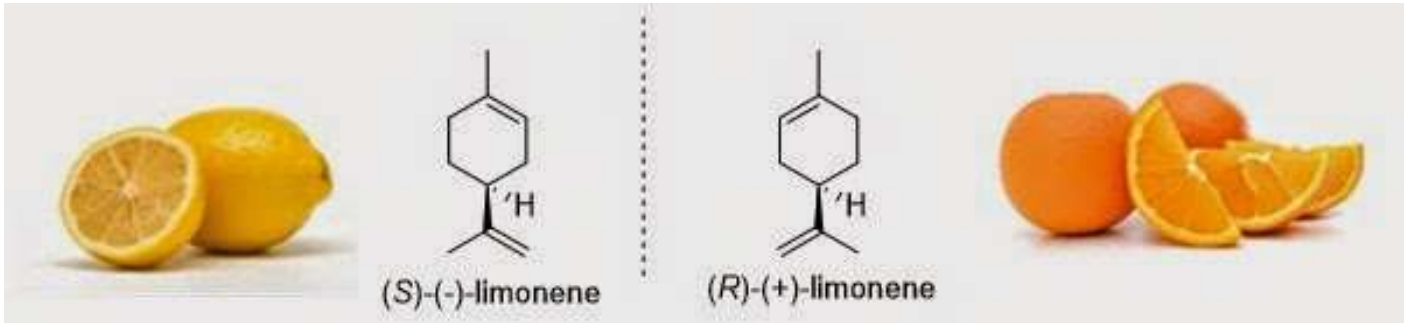

This is not a chiral molecule

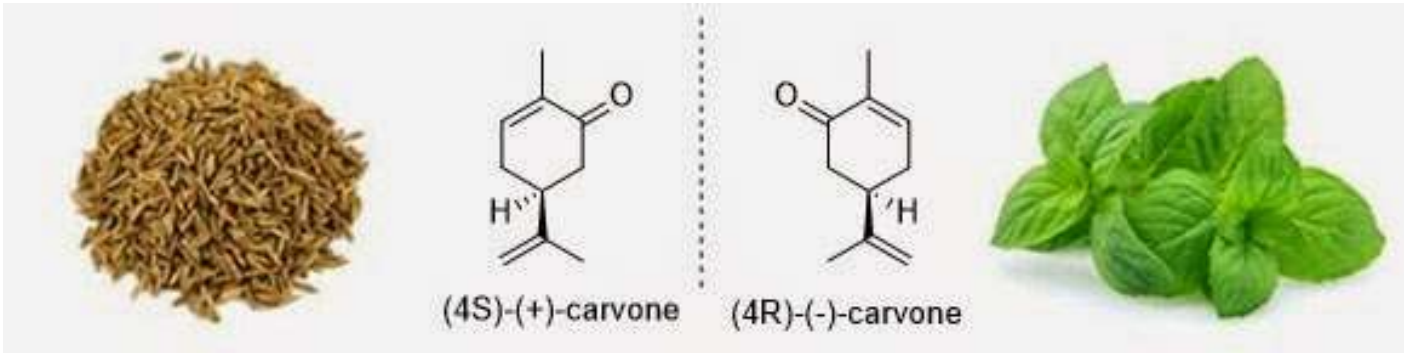



This is a chiral molecule

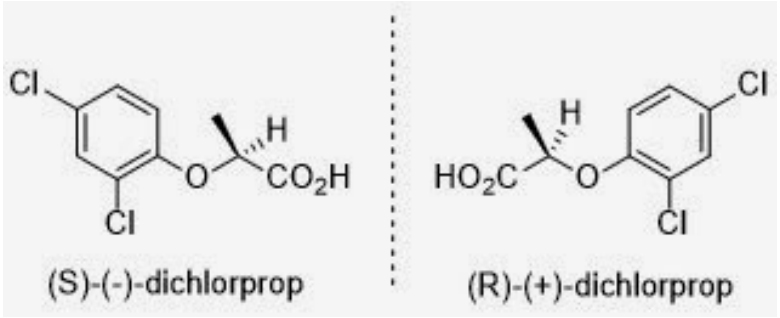
- Two chiral molecules are called enantiomers or isomers
- They are typically named e.g. D and L isomers or R and S, or + and -
- An equal mixture of both enantiomers is called a racemic mixture

Molecules of living organisms (both plants and animals) such as amino acids, peptides, proteins, enzymes, carbohydrates and nucleic acids are chiral.




Herbicide (weed-killer)

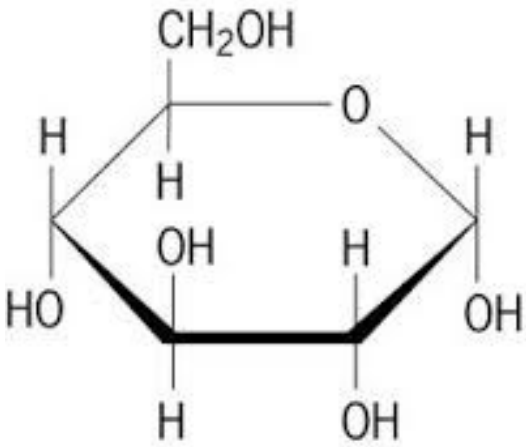



Inactive

https://remotecat.blogspot.com/2014/11/chiral-molecules-in-everyday-life-from.html

**Organic chemistry in the Carbon Cycle: Photosynthesis and Respiration**

Carbohydrates: Sugars, starches and cellulose




glucose

Photosynthesis:

$$6CO_{2\,(g)} + 6H_2O_{\,(l)} + \text{sunlight} = \underset{\text{glucose}}{C_6H_{12}O_6} + 6O_{2\,(g)}$$

Or, more simply, carbohydrates can be represented as $CH_2O$ :

$$CO_{2\,(g)} + H_2O_{\,(l)} + \text{sunlight} = CH_2O + O_{2\,(g)}$$

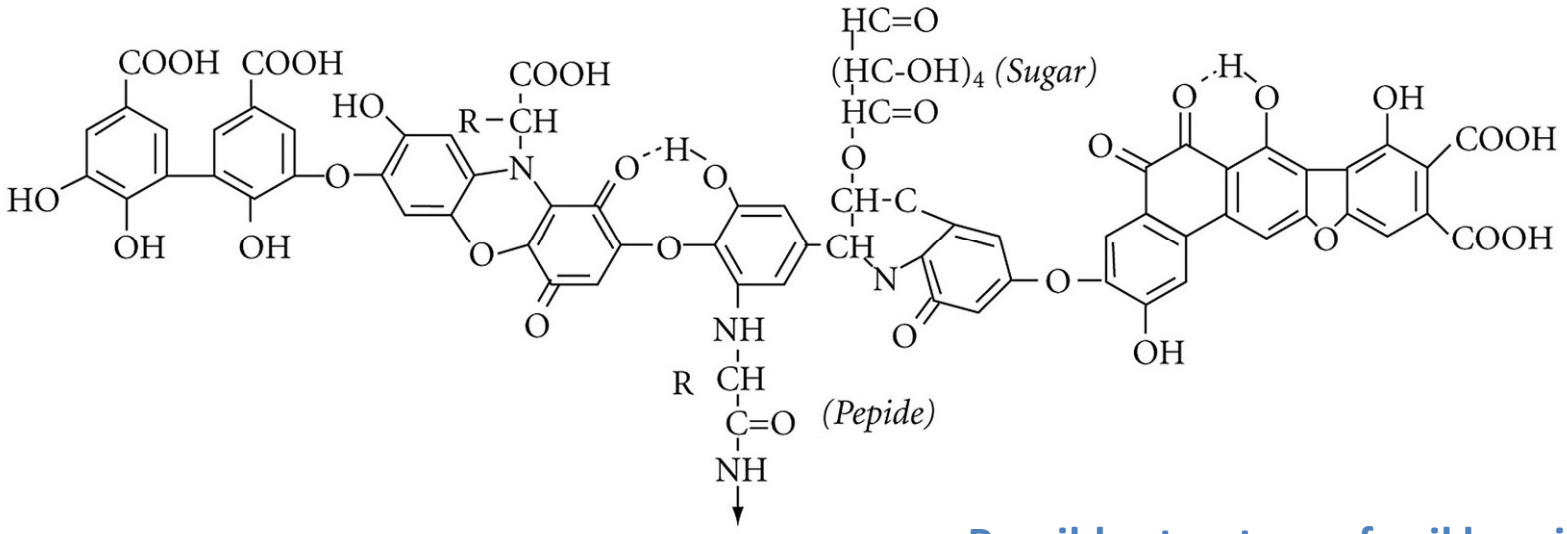
Respiration, reverse of photosynthesis:

$$CH_2O + O_{2\,(g)} = CO_{2\,(g)} + H_2O_{\,(l)} + \text{energy}$$

## Organic matter in sedimentary rocks:

**Soils:** Humic and fulvic acids are produced by biodegradation of organic matter.

**Kerogen:** a mixture of organic compounds that occur in sedimentary rocks. Formed from humic and fulvic acids. High molecular weight organic compounds (can be >1000 a.m.u.). Most abundant organic material in the Earth's crust.




**Possible structure of soil humic acid.**
**Dashed lines represent hydrogen bonds.**

White Fig. 12.23

# Organic chemistry in the Carbon Cycle: Fossil Fuels

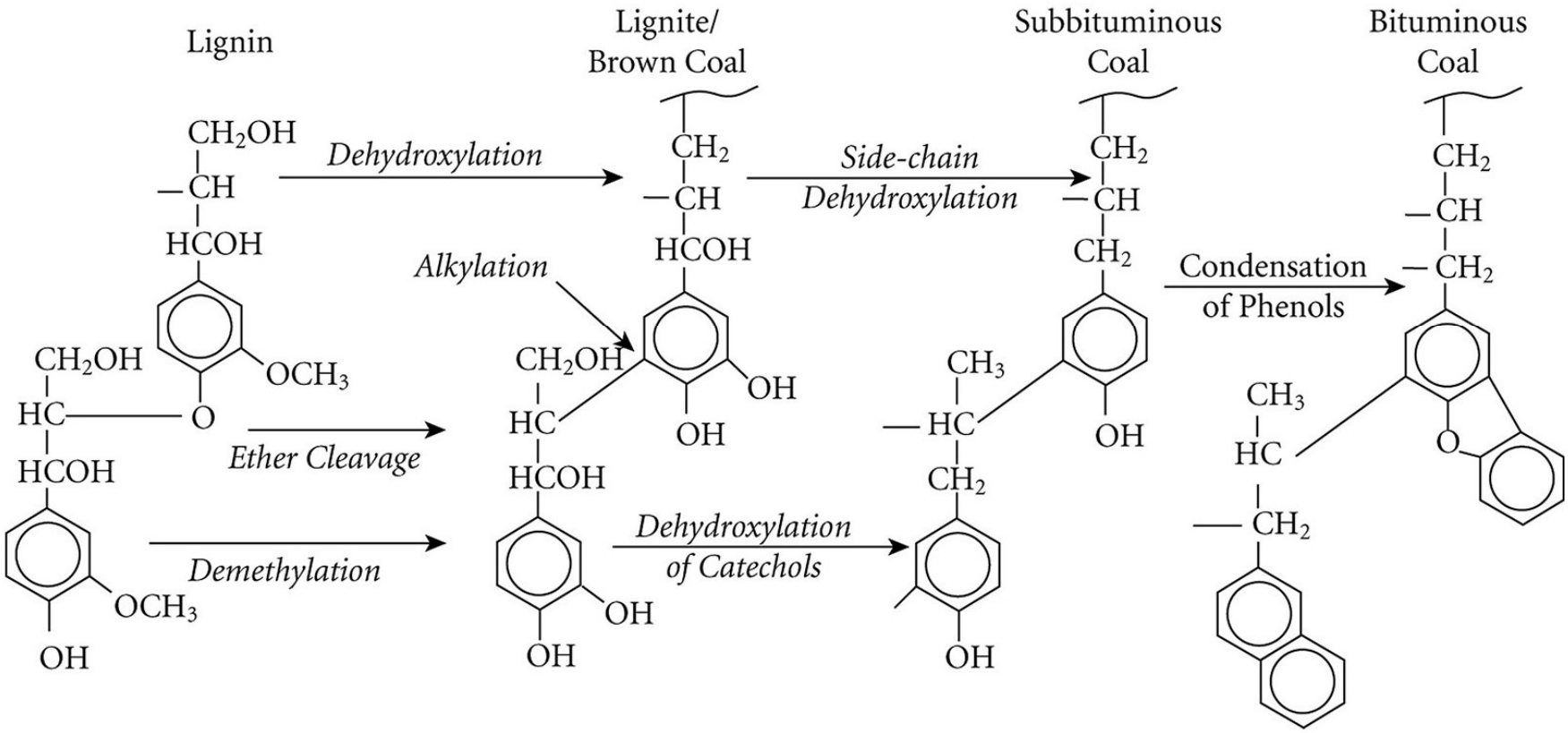
## Formation of coal



*Carboniferous coal swamp*
*M. Parrish*

**Wood** → **Coal**

# Formation of coal




**Lignin - wood**

**Coal**

**Wood** → **Coal**

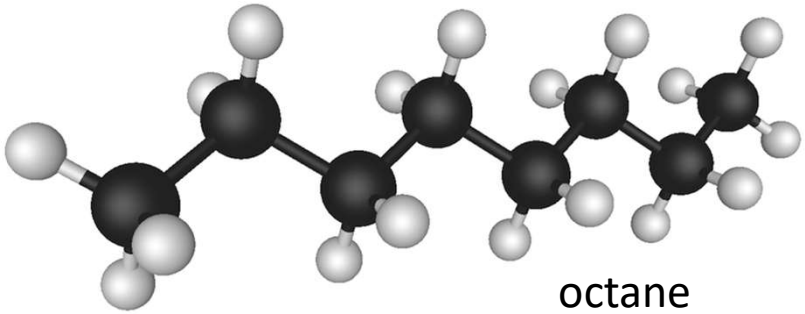
White Fig. 12.41

**Organic matter in sedimentary rocks:**

**Formation of oil and natural gas:**

Breakdown of kerogen forms **oil** at 50-150 °C and **natural gas** at 150-200 °C.

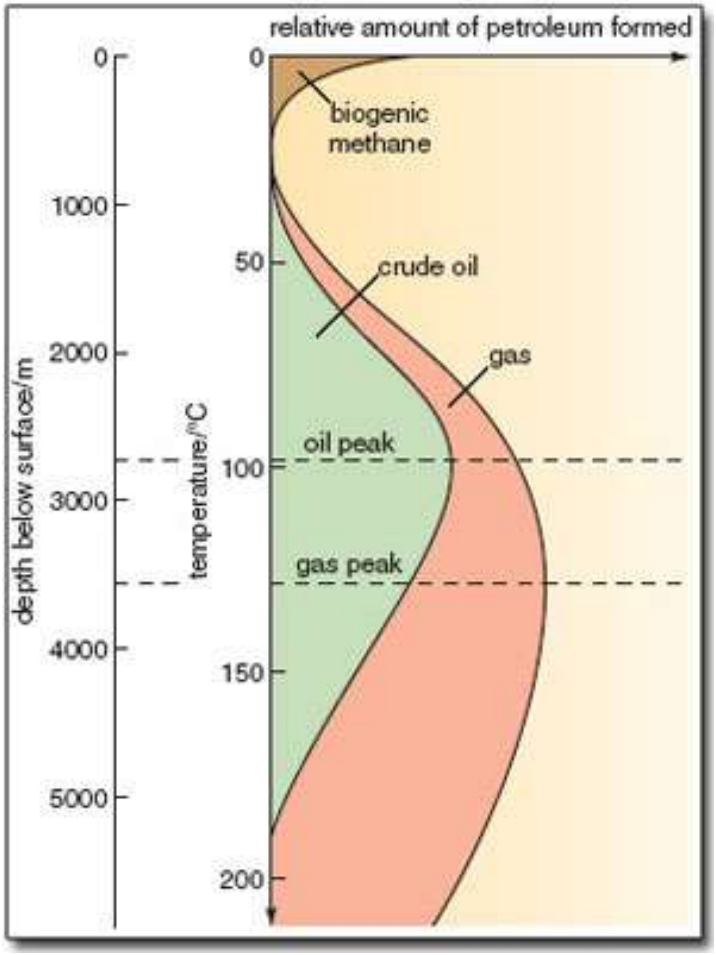
**Oil:** long chain hydrocarbons



octane

**Natural gas:** Methane






Also see White Fig. 12.37

http://www.open.edu/openlearnworks/mod/page/view.php?id=41003

Chemistry of the Earth: Practical 4

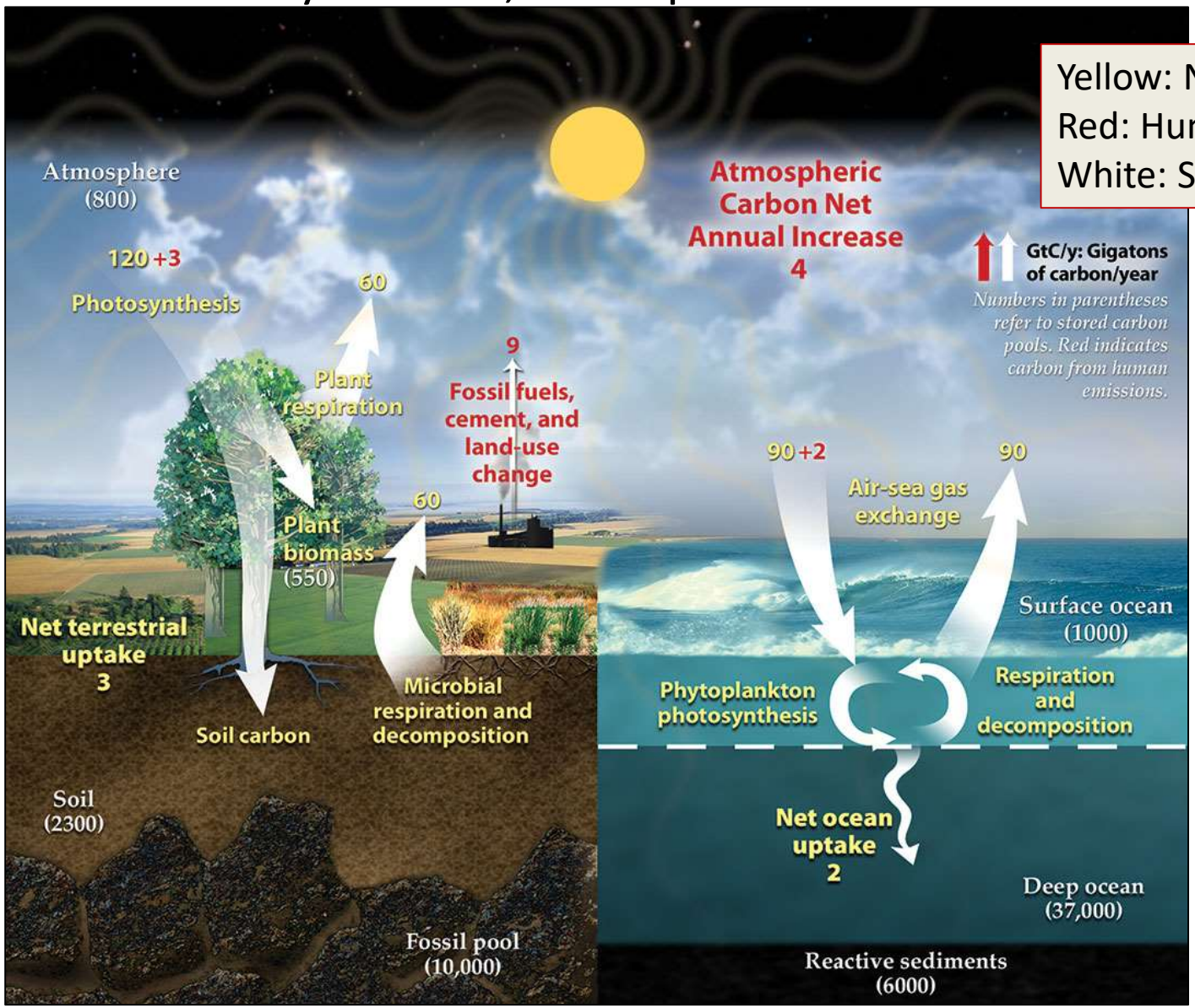
# The Global Carbon Cycle and Organic Chemistry



# Practical 4:
# Organic Chemistry
# and the Carbon Cycle

- **Familiarization with the notation of organic chemistry**
- **Investigation of organic reactions in the petroleum industry, this being related to the anthropogenic (human) contribution to the Carbon Cycle**

# Fast carbon cycle: land, atmosphere and oceans



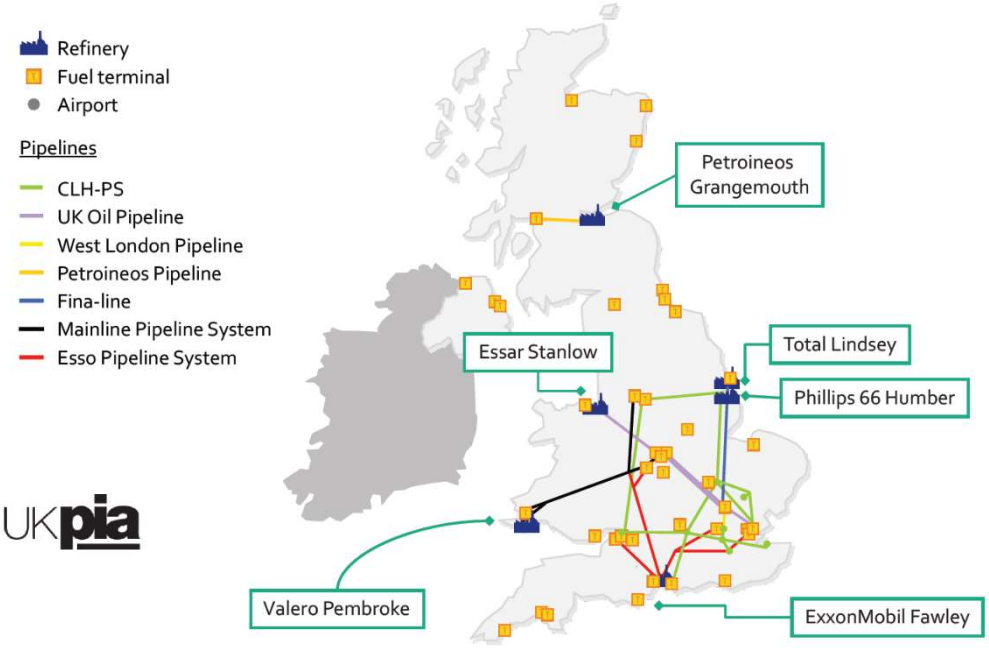

Yellow: Natural fluxes
Red: Human contributions
White: Stored carbon, Gt

**The cycle would be balanced without human intervention**

Figure shows movement of carbon between land, atmosphere and oceans in billions of tons per year (GtC/y).

http://www.carboncycle.org/the-carbon-cycle/

## UK refineries and key product distribution terminals




Note: Indicative summary only – does not contain all locations

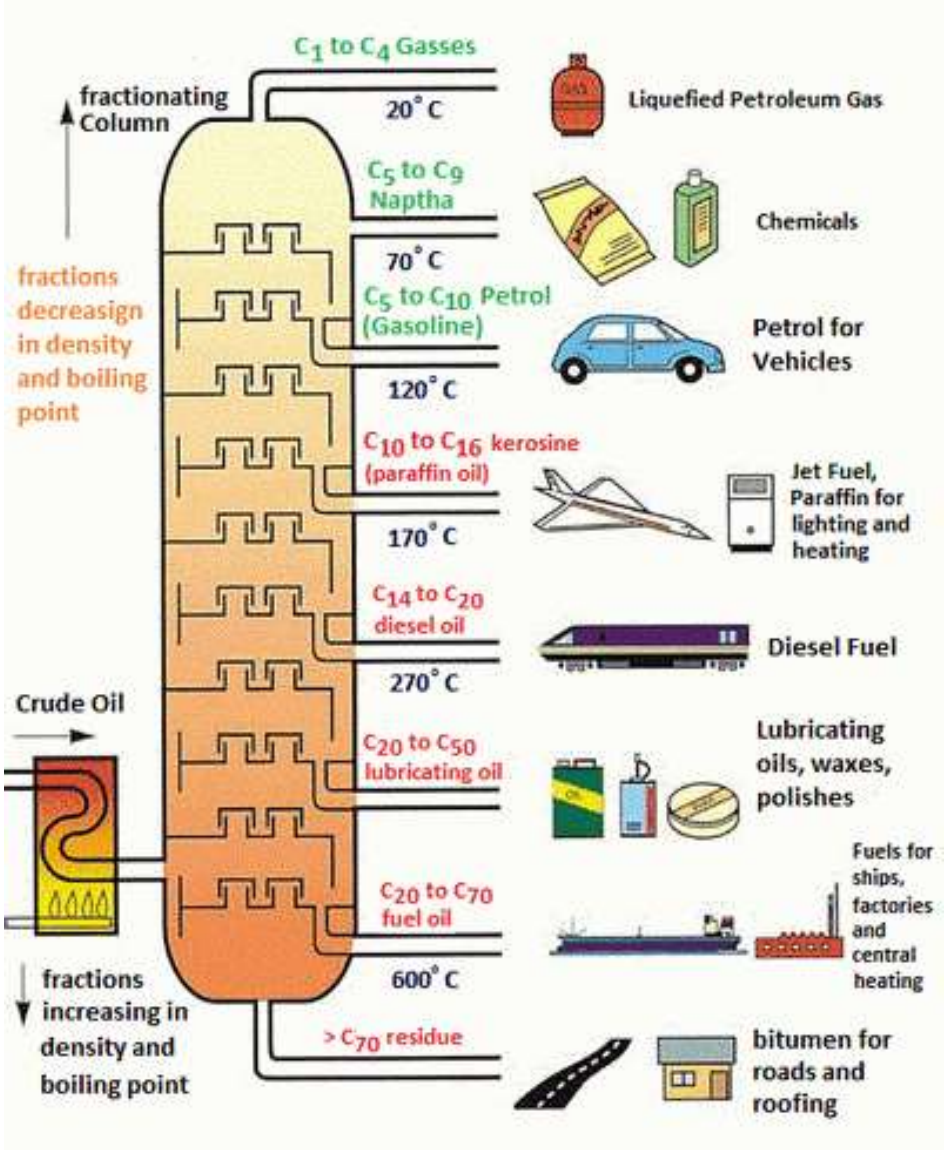
http://www.ukpia.com/industry_information/refining-and-uk-refineries/refineries.aspx

**ESSAR STANLOW,**
**Ellesmere Port**



"Stanlow has a refining capacity of 12 million tonnes per year with a barrel per day capacity of 296,000. Consequently, it is the second largest in the United Kingdom after Fawley Refinery, and produces a sixth of the UK's petrol needs. Stanlow is also a large producer for commodities such as jet fuel and diesel." [Wikipedia]

# OIL REFINERY




https://www.erl.com.bd/refiningprocess.php

